VLSI Devices
Lecture 13

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L13

$V(\phi_s)$?

- Recall that

$$V_{gs} = V_{fb} + \phi_s + \frac{\sqrt{2\epsilon_{si} k_B T N_a}}{C_{ox}} \left[\frac{q\phi_s}{k_B T} + \frac{n_i^2}{N_a^2} \exp\left(\frac{q}{k_B T} (\phi_s - V) \right) \right]^{1/2}$$

Taur, Eq. (5.14)

  - We can rearrange

$$\frac{C_{ox}^2 \left(V_{gs} - V_{fb} - \phi_s \right)^2}{2\epsilon_{si} k_B T N_a} = \frac{q\phi_s}{k_B T} + \frac{n_i^2}{N_a^2} \exp\left(\frac{q}{k_B T} (\phi_s - V) \right)$$

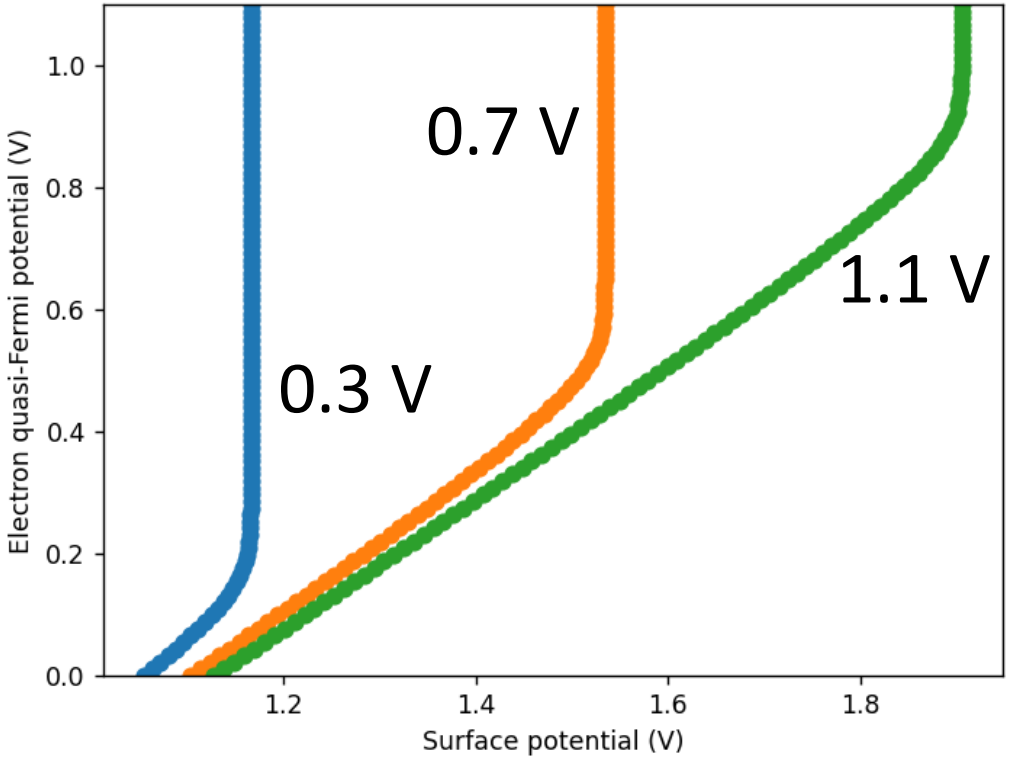
  - Therefore,

$$V = \phi_s - \frac{k_B T}{q} \log \left\{ \frac{N_a^2}{n_i^2} \left[\frac{C_{ox}^2 \left(V_{gs} - V_{fb} - \phi_s \right)^2}{2\epsilon_{si} k_B T N_a} - \frac{q\phi_s}{k_B T} \right] \right\}$$

Taur, Eq. (5.18)

Pao-Sah result

- Consider three gate voltages, V_{gs} = 0.3 V, 0.7 V, and 1.1 V.
  - Draw the electron quasi-Fermi potential (V) as a function of the surface potential (ϕ_s). See Taur, Fig. 5.3.



$\frac{dV}{d\phi_s}$? (1)

- Recall that

$$V = \phi_s - \frac{k_B T}{q} \log\left\{\frac{N_a^2}{n_i^2}\left[\frac{C_{ox}^2\left(V_{gs} - V_{fb} - \phi_s\right)^2}{2\epsilon_{si} k_B T N_a} - \frac{q\phi_s}{k_B T}\right]\right\}$$

Taur, Eq. (5.18)

  - Therefore,

$$\frac{dV}{d\phi_s} = 1 - \frac{k_B T}{q} \frac{-\frac{C_{ox}^2\left(V_{gs} - V_{fb} - \phi_s\right)}{\epsilon_{si} k_B T N_a} - \frac{q}{k_B T}}{\frac{C_{ox}^2\left(V_{gs} - V_{fb} - \phi_s\right)^2}{2\epsilon_{si} k_B T N_a} - \frac{q\phi_s}{k_B T}}$$

$\frac{dV}{d\phi_s}$? (2)

- Simple rearrange yields

$$\frac{dV}{d\phi_s} = 1 + \frac{2k_BT}{q}\frac{C_{ox}^2\left(V_{gs} - V_{fb} - \phi_s\right) + \epsilon_{si}qN_a}{C_{ox}^2\left(V_{gs} - V_{fb} - \phi_s\right)^2 - 2\epsilon_{si}qN_a\phi_s}$$

Taur, Eq. (5.19)

  - It is still very complicated…

Integrand

- When multiplied with $-Q_i(\phi_s)$,

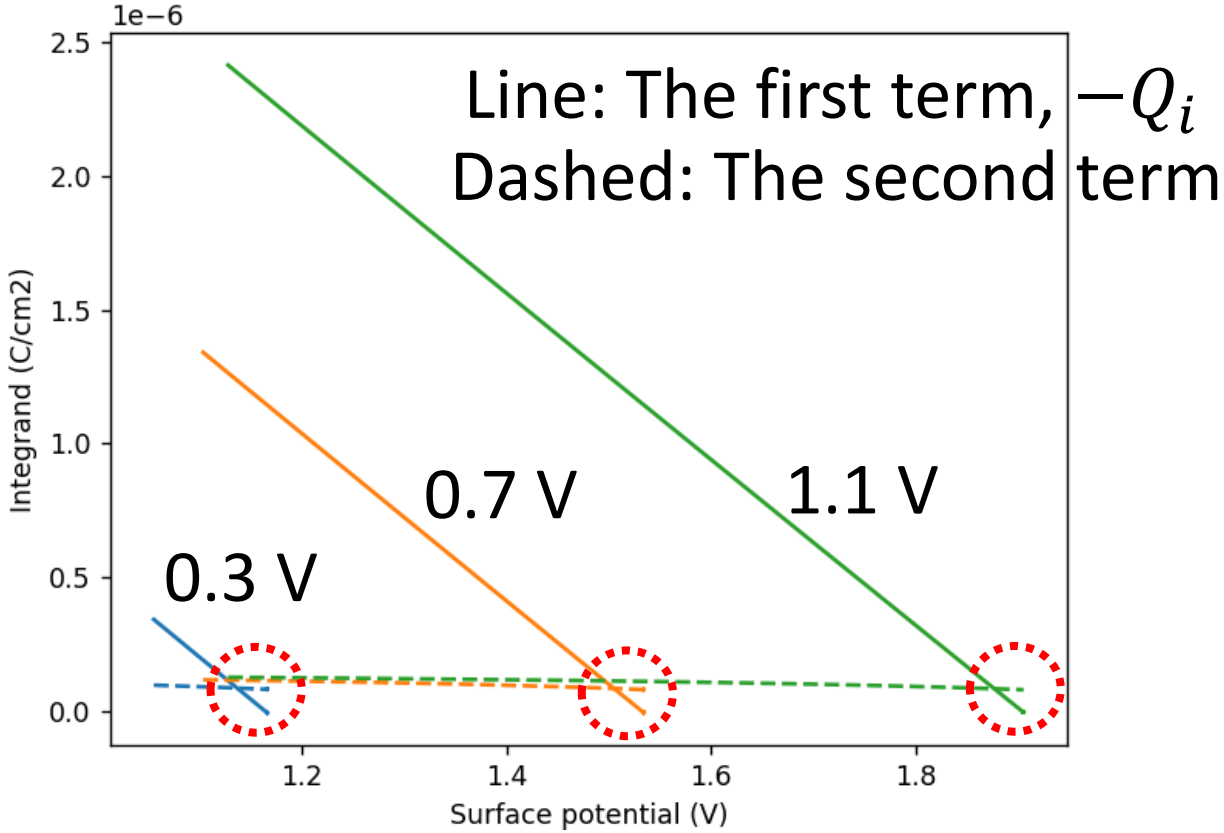
$$\left(-Q_i(\phi_s)\right)\frac{dV}{d\phi_s}$$

$$= -Q_i(\phi_s) + \frac{2k_BT}{q}\frac{C_{ox}^2\left(V_{gs} - V_{fb} - \phi_s\right) + \epsilon_{si}qN_a}{C_{ox}\left(V_{gs} - V_{fb} - \phi_s\right) + \sqrt{2\epsilon_{si}qN_a\phi_s}}$$

  - The second term is still very complicated…
  - *Is it really important?*

Comparison between two terms

- Let's draw two terms.
  - Assume that N_a = 10^{18} cm^{-3}, t_{ox} = 1.2 nm, and V_{fb} = -1.08 V.
  - The second term is small.
  - It is meaningful only when $Q_i \approx 0$.
  - This is corresponding to $C_{ox}(V_{gs} - V_{fb} - \phi_s) = \sqrt{2\epsilon_{si} q N_a \phi_s}$.



Integrand, again

- Within this condition,

$$\left(-Q_i(\phi_s)\right)\frac{dV}{d\phi_s} \approx -Q_i(\phi_s) + \frac{k_BT}{q}\frac{C_{ox}\sqrt{2\epsilon_{si}qN_a\phi_s} + \epsilon_{si}qN_a}{\sqrt{2\epsilon_{si}qN_a\phi_s}}$$

$$= -Q_i(\phi_s) + \frac{k_BT}{q}C_{ox} + \frac{k_BT}{q}\frac{\sqrt{2\epsilon_{si}qN_a}}{2\sqrt{\phi_s}}$$

Its integration yields $\frac{k_BT}{q}C_{ox}\phi_s$.

Its integration yields $\frac{k_BT}{q}\sqrt{2\epsilon_{si}qN_a\phi_s}$.

Drain current

- Using the previous approximation,
  - We can obtain the following expression:

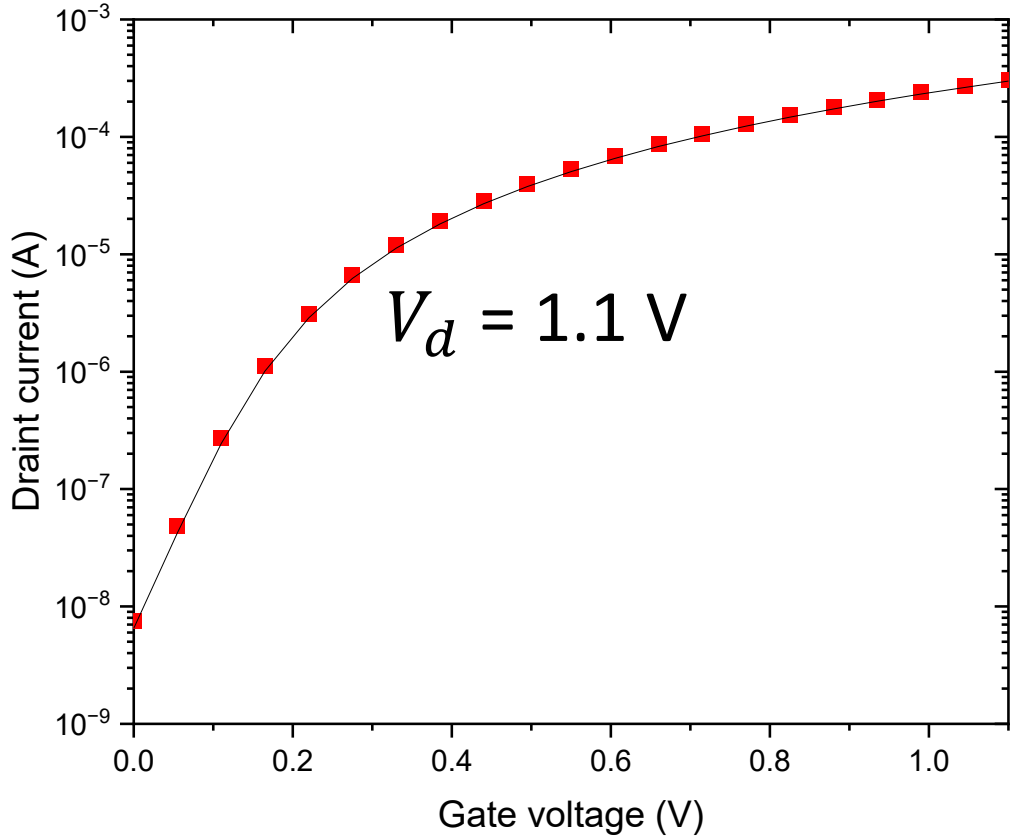
$$I_d = \mu_{eff}\frac{W}{L}\left\{C_{ox}\left(V_{gs} - V_{fb} + \frac{k_B T}{q}\right)\phi_s - \frac{1}{2}C_{ox}\phi_s^2 - \frac{2}{3}\sqrt{2\epsilon_{si}qN_a}\phi_s^{1.5} + \frac{k_B T}{q}\sqrt{2\epsilon_{si}qN_a\phi_s}\right\}\Bigg|_{\phi_{s,s}}^{\phi_{s,d}}$$

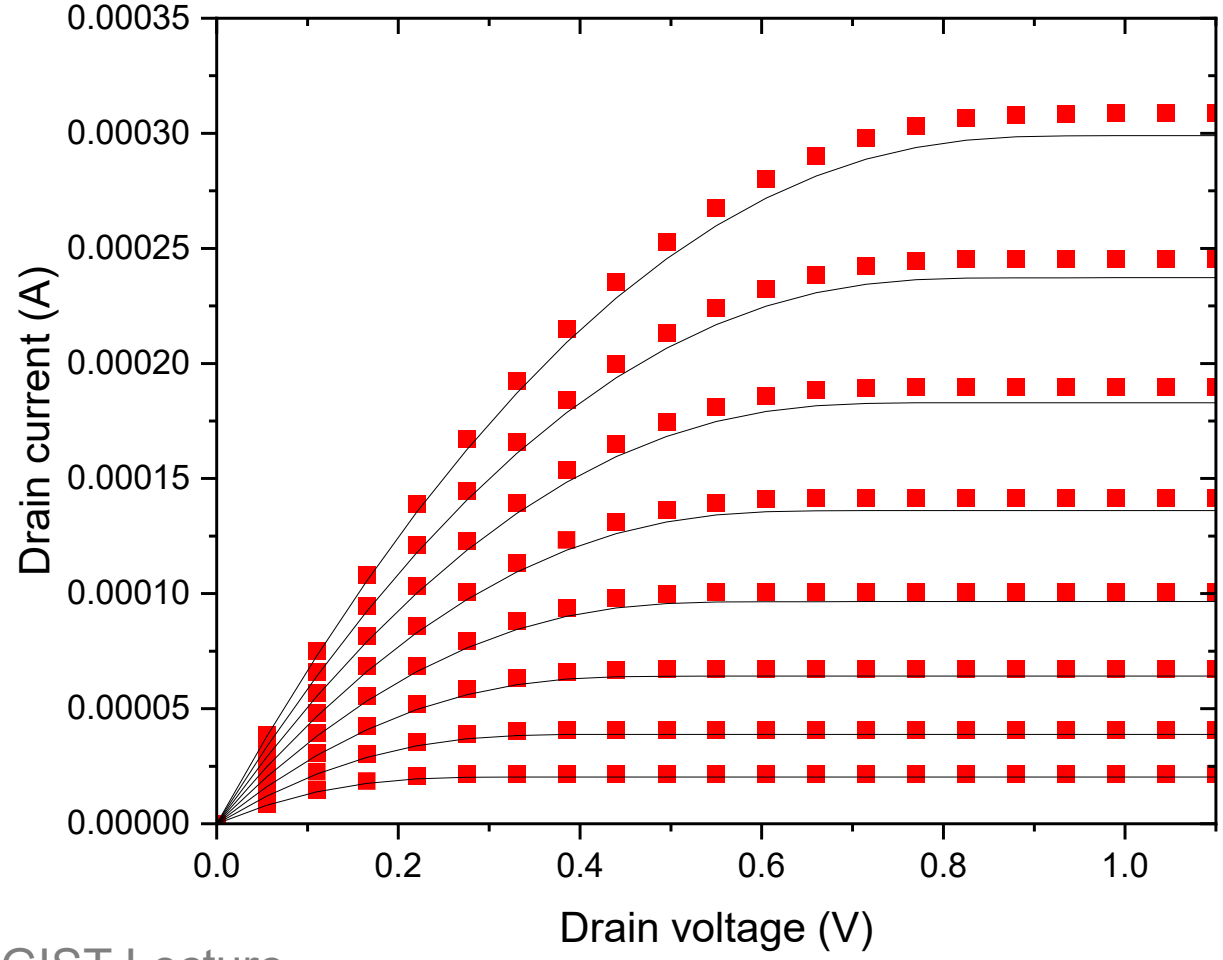
Taur, Eq. (5.21)

  - Only with $\phi_{s,s}$ and $\phi_{s,d}$, we can evaluate the drain current.

IV curves, again

- Output curves by two models (Pao-Sah & charge-sheet)
  - Symbols: Pao-Sah
  - Lines: Charge-sheet





Regional approximations

- After the onset of inversion but *before saturation*,
  - The surface potential, $\phi_s(y)$, can be approximated by

$$\phi(0,y) = V(y) + 2\phi_B$$

Taur, Eq. (5.3)

  - It means that

$$\phi_{s,s} = 2\phi_B$$
$$\phi_{s,d} = 2\phi_B + V_{ds}$$

  - In this case, $\frac{dV}{d\phi_s} = 1$. We must calculate the following term for $\phi_{s,d}$:

$$C_{ox}\left(V_{gs} - V_{fb}\right)(2\phi_B + V_{ds}) - \frac{1}{2}C_{ox}(2\phi_B + V_{ds})^2 - \frac{2}{3}\sqrt{2\epsilon_{si}qN_a}(2\phi_B + V_{ds})^{1.5}$$

A simpler form of I_d

- By taking the difference, we can find a simpler form:

$$I_d = \mu_{eff}\frac{W}{L}\left\{C_{ox}\left(V_{gs} - V_{fb} - 2\phi_B - \frac{1}{2}V_{ds}\right)V_{ds} - \frac{2}{3}\sqrt{2\epsilon_{si}qN_a}\left[(2\phi_B + V_{ds})^{1.5} - (2\phi_B)^{1.5}\right]\right\}$$

Taur, Eq. (5.22)

  - For a given V_{gs}, I_d first increases linearly with V_{ds}, then gradually levels off to a saturated value.

Linear (triode) region

- When V_{ds} is small, we may keep only up to the first order.

$$I_d = \mu_{eff}\frac{W}{L}\left\{C_{ox}\left(V_{gs}-V_{fb}-2\phi_B\right)V_{ds}-\frac{2}{3}\sqrt{2\epsilon_{si}qN_a}\left[\frac{3}{2}(2\phi_B)^{0.5}V_{ds}\right]\right\}$$

$$= \mu_{eff}\frac{W}{L}C_{ox}\left(V_{gs}-V_{fb}-2\phi_B-\frac{\sqrt{4\epsilon_{si}qN_a\phi_B}}{C_{ox}}\right)V_{ds}$$

$$= \mu_{eff}\frac{W}{L}C_{ox}\left(V_{gs}-V_t\right)V_{ds}$$

Taur, Eq. (5.23)

- The threshold voltage, V_t, is given by

$$V_t = V_{fb}+2\phi_B+\frac{\sqrt{4\epsilon_{si}qN_a\phi_B}}{C_{ox}}$$

Taur, Eq. (5.24)

- It is the gate voltage when the surface potential reaches at $2\phi_B$.

Parabolic region

- We must keep up to the second order.

$$I_d = \mu_{eff}\frac{W}{L}\left\{C_{ox}\left(V_{gs} - V_{fb} - 2\phi_B - \frac{1}{2}V_{ds}\right)V_{ds} - \frac{2}{3}\sqrt{2\epsilon_{si}qN_a}\left[\frac{3}{2}(2\phi_B)^{0.5}V_{ds} + \frac{3}{8}(2\phi_B)^{-0.5}V_{ds}^2\right]\right\}$$

$$= \mu_{eff}\frac{W}{L}\left\{C_{ox}\left(V_{gs} - V_t - \frac{1}{2}V_{ds}\right)V_{ds} - \frac{1}{4}\sqrt{2\epsilon_{si}qN_a}\left[(2\phi_B)^{-0.5}V_{ds}^2\right]\right\}$$

$$= \mu_{eff}\frac{W}{L}\left\{C_{ox}(V_{gs} - V_t)V_{ds} - \frac{1}{2}C_{ox}\left[1 + \frac{\sqrt{\epsilon_{si}qN_a/(4\phi_B)}}{C_{ox}}\right]V_{ds}^2\right\}$$

Taur, Eq. (5.25)

Thank you!